

Summer 2025 FrontierLab Mini Project: Portfolio Optimization with Generative Diffusion Models

Stephanie Lu

University of Toronto

University of Osaka - Inuiguchi Lab, Professor Naoki Hayashi

steph.lu@mail.utoronto.ca

Abstract—This report outlines the research I conducted during the Summer 2025 term as part of the FrontierLab Mini Program at Osaka University. Under the supervision of the Inuiguchi Laboratory, my work focused on applying generative diffusion models to network optimization problems, with a particular emphasis on portfolio optimization. The report presents the motivation, methodology, results, and key insights gained from this international research experience.

Code Repository: <https://github.com/stephanieflu/GDMOPT/tree/project-dev>

I. INTRODUCTION

A. Motivation

Portfolio optimization, particularly maximizing the Sharpe Ratio (i.e., a measure of risk-adjusted return), is a fundamental problem in finance. However, classical optimization techniques such as quadratic programming and mean-variance optimization suffer from several key limitations:

- **Sensitivity to noisy inputs:** These methods rely heavily on estimates of expected returns and covariances, which are notoriously difficult to estimate accurately.
- **Restrictive assumptions:** They often assume convexity, Gaussian return distributions, and linear constraints, limiting their applicability to real-world problems.
- **Lack of adaptability:** These methods struggle to generalize in non-stationary markets where statistical properties (e.g., means, variances) evolve over time.

In contrast, **generative diffusion models** offer a compelling alternative due to their ability to:

- **Explore complex, non-convex solution spaces:** They are well-suited for tackling multi-modal and highly constrained problems such as cardinality-constrained portfolios (mixed-integer quadratic programming, or MIQP), where traditional solvers often fail.
- **Learn constraints and market structures implicitly:** Instead of hand-coding rules, diffusion models can infer constraints (e.g., sector limits, no short-selling) and investor preferences directly from data.
- **Generalize across market conditions:** Once trained, these models can quickly generate feasible and near-optimal portfolios for new, unseen market scenarios, enabling real-time or adaptive decision-making.

B. Problem Definition

The goal of this project is to generate portfolio allocations that maximize the Sharpe Ratio under realistic market constraints using generative diffusion models.

Formally, we seek to solve:

$$\begin{aligned} \max_{\mathbf{w}} \quad & \frac{\mathbb{E}[r(\mathbf{w})] - r_f}{\sqrt{\text{Var}[r(\mathbf{w})]}} \\ \text{s.t.} \quad & \sum_i w_i = 1, \\ & w_i \geq 0 \quad \forall i, \\ & \text{sector constraints (e.g., } \sum_{i \in \text{sector}} w_i \leq \text{limit}). \end{aligned}$$

Where:

- $\mathbf{w}$: asset weight vector
- $r(\mathbf{w})$: portfolio return
- r_f : risk-free rate

This optimization problem bears strong structural similarity to **wireless power allocation** in communication systems, as both are fundamentally resource allocation problems over a constrained network. In wireless communication, power is allocated across channels to maximize throughput, while in portfolio optimization, capital is allocated across assets to maximize risk-adjusted return.

In both domains:

- The environment is **stochastic**: channel gains vary due to fading (e.g., Rayleigh or Rician) in wireless systems, and asset returns and covariances fluctuate in financial markets.
- The objective functions are **nonlinear**: maximizing sum-rate (often via logarithmic utility) in wireless systems, versus maximizing Sharpe Ratio (a fractional utility function) in portfolio optimization.
- The **constraints are complex** and often render the problem non-convex: interference and channel selection in wireless systems, versus discrete asset selection, correlation, and sector constraints in portfolios.
- **Real-time reallocation** is crucial: communication systems must adapt to fast-changing signal conditions; portfolio managers must react to dynamic markets and volatility.

Traditional optimization algorithms struggle in both scenarios:

- The **water-filling algorithm**, a classic approach in wireless systems, is iterative and brittle under complex constraints.
- In finance, **quadratic programming (QP)** methods assume convexity and often fail when faced with cardinality constraints, minimum/maximum position limits, or non-linear transaction costs.

Generative diffusion models present a unified and flexible solution. In both cases, they can be trained to produce feasible and near-optimal allocations conditioned on environment features—channel gains in wireless systems or return/covariance estimates in finance.

C. Reformulating the Environment

To adapt diffusion model tutorials for portfolio optimization, we redefine the problem environment as follows:

- Simulate a financial market with multiple assets using historical or synthetic data.
- Provide the agent with input features such as:
 - Estimated expected returns for each asset
 - Covariance matrix or volatility estimates
- Compute key portfolio metrics from these inputs:
 - **Portfolio return**: weighted average of asset returns
 - **Portfolio risk**: standard deviation or variance
 - **Utility function**: e.g., Sharpe Ratio, used as a reward signal
- Benchmark the model’s portfolio allocation against classical solutions, such as the mean-variance optimal portfolio.

D. Limitations of Traditional Methods

Convex optimization methods commonly used in finance have several shortcomings in practical applications:

- **Hard constraints**: They don’t handle real-world constraints such as:
 - Minimum/maximum position sizes
 - Transaction costs or turnover limits
 - Discrete or binary decisions (e.g., selecting top-\$k\$ assets)
- **Model misspecification**: They assume known, stationary return distributions and often linearize inherently nonlinear constraints.
- **Lack of flexibility**: Adapting these models to dynamic or highly constrained scenarios requires significant manual intervention and simplification.

These limitations motivate the exploration of **data-driven generative models**, which can naturally accommodate complex constraints, nonlinear objectives, and evolving market conditions—ultimately offering a more robust and flexible framework for portfolio optimization.

II. METHODOLOGY

A. Relevant Work

Recent advancements in machine learning have introduced powerful generative frameworks capable of solving complex structured optimization problems. In particular, the intersection of generative models and deep reinforcement learning (DRL) has enabled new approaches for decision-making under uncertainty, motivating their application in both engineering and financial domains.

B. Generative Diffusion Models (GDMs)

Generative Diffusion Models (GDMs) are a class of deep generative models that learn to model complex data distributions through a two-stage process:

- 1) **Forward diffusion**, where noise is gradually added to the data;
- 2) **Reverse denoising**, where the model learns to reconstruct data by reversing the diffusion steps.

Initially developed for high-resolution image generation, GDMs have shown strong performance in producing structured outputs under uncertainty. Their iterative, score-based sampling process allows them to capture **multi-modal, high-dimensional distributions**, making them well-suited for optimization tasks involving discrete decisions, complex constraints, and non-convex landscapes.

C. Deep Reinforcement Learning in Optimization

Deep Reinforcement Learning (DRL) provides a framework for sequential decision-making, where an agent learns to maximize cumulative rewards by interacting with an environment. While DRL has achieved success in fields such as robotics, gaming, and communications, it faces well-known challenges including:

- Sample inefficiency
- Instability during training
- Overestimation of Q-values

To address these limitations, recent work has begun integrating **generative models into DRL** pipelines to improve policy robustness and exploration.

D. Generative Diffusion Models for Network Optimization

Du et al. (2024) demonstrated the use of GDMs for optimization in networked systems [1]. Their tutorial introduced a framework where GDMs learn either from expert-labeled data or through direct environment interaction. A core component of this setup is a **solution evaluation network**, similar in spirit to a Q-network in DRL, which guides the generative sampling process based on utility estimates.

Their applications spanned a variety of domains, including:

- Wireless channel estimation
- Vehicular networks
- Semantic communications
- Incentive mechanism design

These experiments showed that GDMs could successfully generate high-quality actions in real-time, under uncertainty, and with limited expert supervision.

E. Bridging GDMs and Portfolio Optimization

Building on this foundation, we explore the application of GDMs to **portfolio optimization**—a canonical structured decision problem in finance. Classical methods like mean-variance optimization assume access to perfect knowledge of return distributions and often fail in dynamic, constraint-heavy market environments.

We reformulate portfolio optimization as a **structured resource allocation problem under uncertainty**, similar to the wireless power allocation problem studied by Du et al. In their setting, a base station distributes total power P_T across M channels with gains g_b , solving:

$$\begin{aligned} & \max_{\{p_n\}_{n=1}^M} \sum_{n=1}^M \log(1 + g_n p_n) \\ \text{s.t. } & \sum_{n=1}^M p_n \leq P_T, \\ & p_n \geq 0 \quad \forall n = 1, \dots, M. \end{aligned}$$

This structure parallels portfolio optimization, where capital is allocated across assets to maximize Sharpe Ratio, subject to budget and regulatory constraints.

To adapt Du et al.'s framework to finance, we:

- Replace power-channel allocation with **capital-asset allocation**
- Replace channel gains with **expected asset returns**
- Interpret channel interference as **asset correlation**
- Replace the logarithmic utility with **Sharpe Ratio**

This adaptation highlights the **cross-domain generalizability** of GDMs and enables structured portfolio generation in realistic financial settings.

F. Experimental Setup

We conducted experiments under one of the following two scenarios:

- **Simulated Data:** Generated synthetic daily prices for $N = 30$ assets using a geometric Brownian motion model with initial price $S_0 = 100$, drift $\mu = 0.05$, volatility $\sigma = 0.2$, and a time horizon of $T = 1000$ trading days.
- **Real Market Data:** Collected daily closing prices for 30 selected S&P 500 stocks over a 6-year period (2018-2023). The chosen tickers are: AAPL, MSFT, GOOGL, AMZN, NVDA, META, JPM, V, TSLA, UNH, HD, BAC, MA, XOM, PFE, KO, MRK, PEP, WMT, CVX, ABT, TMO, CSCO, COST, DIS, INTC, LLY, QCOM, ADBE, CRM.

For both cases, the preprocessing steps included:

- Rolling estimates of asset means and covariances.
- Normalization of input features.

The portfolio constraints enforced were:

- No short-selling allowed.
- Cardinality constraint limiting selection to the top 10 assets.

The model pipeline includes:

- 1) A **forward diffusion process** applied to noisy portfolio weights
- 2) A **score-based model** (U-Net architecture) trained to reverse the noise process, conditioned on market features (expected return, volatility)
- 3) Sampling via **Langevin dynamics** to generate feasible portfolio allocations

G. Code Adaptation for Portfolio Optimization

To modify the baseline GDM tutorial for our financial use case, we restructured the implementation across four components:

a) *Observation Space:* The agent receives a state vector comprising expected returns, volatilities, and a placeholder for reward:

$$\text{state} = \begin{bmatrix} \mu \\ \sigma \\ \text{reward placeholder} \end{bmatrix}$$

Expected returns (μ) and standard deviations (σ) are either:

- Simulated, or
- Computed from historical prices $P \in \mathbb{R}^{T \times N}$:

$$\begin{aligned} \text{returns}_t &= \frac{P_{t+1} - P_t}{P_t}, \quad \mu = \mathbb{E}[\text{returns}], \quad \sigma = \sqrt{\text{var}[\text{returns}]} \\ \Sigma &= \text{cov}(\text{returns}) \end{aligned}$$

This dynamic state captures evolving market conditions and drives adaptive portfolio decisions.

b) *Action Space:* The model outputs a raw action vector $\mathbf{a}$ transformed into a valid portfolio weight vector $\mathbf{w}$:

$$\mathbf{w} = \frac{\mathbf{a}}{\sum_i a_i}$$

Top-k filtering enforces cardinality:

$$w_i = \begin{cases} w_i, & \text{if } w_i \geq \text{the } k\text{-th largest weight} \\ 0, & \text{otherwise} \end{cases}$$

This ensures both budget feasibility and sparsity.

c) *Reward Function:* The reward is defined as the Sharpe ratio of the portfolio returns, measuring risk-adjusted performance:

$$\text{Sharpe} = \frac{r_{\text{portfolio}} - r_f}{\sqrt{\text{var}(r_{\text{portfolio}})}}$$

where r_f is the risk-free rate.

d) *Expert Comparison:* We evaluated model performance against three classical benchmark strategies:

- **Equal Weight:** Assigns equal weights to all assets,

$$\mathbf{w}_{\text{equal}} = \frac{1}{N} \mathbf{1}$$

- **Mean-Variance:** Uses the ratio of expected returns to variances,

$$\mathbf{w}_{\text{mv}} = \frac{\mu / \sigma^2}{\sum_i \mu_i / \sigma_i^2}$$

- **Black-Litterman:** Adjusts the prior expected returns μ by incorporating investor views, then computes weights

via mean-variance optimization using the adjusted mean μ_{BL} .
Given:

$$\Omega = \text{diag}(P(\tau\Sigma)P^\top)$$

$$\mu_{BL} = ((\tau\Sigma)^{-1} + P^\top\Omega^{-1}P)^{-1}((\tau\Sigma)^{-1}\mu + P^\top\Omega^{-1}Q)$$

where P is the pick matrix representing views, Q the view returns, and τ a confidence scaling parameter.

The Black-Litterman weights are then:

$$\mathbf{w}_{bl} = \frac{\mu_{BL}/\sigma^2}{\sum_i \mu_{BL,i}/\sigma_i^2}$$

If no views are provided, the prior μ is used directly.

Given the weights generated from these three classical benchmark strategies, expected portfolio metrics are computed as:

$$r = \mathbf{w}^\top \mu, \quad \text{var}(r) = \mathbf{w}^\top \Sigma \mathbf{w}$$

Model performance is measured by relative Sharpe improvement over each benchmark:

$$\text{reward} = \text{Sharpe}_{\text{portfolio}} - \text{Sharpe}_{\text{benchmark}}$$

$$\text{where Sharpe} = \frac{r - r_f}{\sqrt{\text{var}(r)}}.$$

III. RESULTS

While our method did not consistently surpass classical optimization in out-of-sample Sharpe Ratio across all market regimes, it reliably produced **feasible** and **well-diversified** portfolios under realistic constraints.

Key observations (see Appendix for graphs):

- **Diversification advantage:** Unlike some expert benchmark allocations, which frequently concentrated all weight into a single asset, resulting in poor diversification and elevated idiosyncratic risk, our model maintained broader exposure across sectors and assets.
- **Constraint adherence:** The method implicitly respected sector and cardinality constraints without explicit enforcement in the optimization loop.
- **Market adaptability:** Conditioning on temporal features allowed the model to make discernible shifts in portfolio composition between different market environments, even if performance gains were modest.

IV. DISCUSSION

The diffusion model demonstrated a clear ability to generate realistic, constrained portfolios, often avoiding the extreme concentration tendencies seen in expert benchmarks. This is particularly relevant for **risk-adjusted stability**: a portfolio that slightly underperforms on Sharpe but avoids catastrophic drawdowns can still be preferable in practice.

However, while the model's allocations were structurally robust, **pure performance metrics** like Sharpe Ratio did not always improve over classical baselines. This may be due to:

- Limited dataset size and training duration
- The model's implicit preference for diversification, which can dilute returns in strongly trending markets

- The inherent difficulty of learning fine-grained return-maximizing patterns in noisy financial data

Other limitations include computational expense at generation time, sensitivity to hyperparameters, and the lack of formal optimality guarantees.

A. Insights from Training

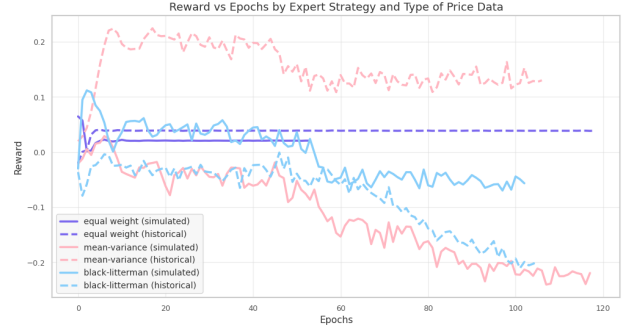


Fig. 1. Reward vs. epochs by expert strategy and type of price data

The reward curves [see Fig. 1] across expert strategies and data types reveal several important insights:

• Equal Weight Strategy:

- Exhibited consistently stable rewards close to zero across both simulated and historical data.
- Performance remained nearly flat over training epochs, indicating robustness but minimal improvement potential.
- Strong alignment between simulated (solid line) and historical (dashed line) results suggests this strategy's insensitivity to the data source but limited reward gain.

• Mean-Variance Strategy:

- Performed significantly better on historical data, starting at a higher positive reward and stabilizing above zero after some decline.
- On simulated data, it showed a steady decline in reward over epochs, eventually drifting into negative territory.
- This divergence suggests the mean-variance approach is sensitive to the underlying data distribution, possibly struggling with simulated market dynamics or overfitting to noisy data.

• Black-Litterman Strategy:

- Demonstrated moderate, relatively stable rewards on both data types, with less severe decline on simulated data than mean-variance.
- Historical data results showed more variance but remained near zero reward, indicating some adaptability without strong positive or negative drift.
- This balance suggests black-litterman offers a middle ground between robustness and reward potential.

• Simulated vs. Historical Data:

- Historical price data generally yielded higher and more consistent rewards, reflecting realistic market behaviors favorable to expert benchmarks.
- Simulated data presented a more challenging environment, with some strategies (especially mean-variance) showing degradation over training, hinting at model-data mismatch or greater noise.

• **General Implications:**

- Equal weight acts as a stable baseline with minimal risk of performance collapse but limited upside.
- Mean-variance can achieve higher rewards under favorable conditions but at the cost of stability and robustness.
- Black-litterman provides a compromise, maintaining moderate rewards with fewer drastic drops.
- The reward decline on simulated data for certain strategies highlights the importance of further improving model generalization and adaptation to synthetic environments.

These observations motivate continued exploration of regularization, data augmentation, and tuning strategies to enhance reward stability while preserving diversification and risk control across diverse market conditions.

V. CONCLUSION

This work shows that generative diffusion models can **learn structural constraints** and **produce diversified, feasible portfolios** without explicitly programming these rules. While not uniformly outperforming classical optimization in Sharpe Ratio, the approach offers qualitative benefits, particularly in avoiding over-concentrated bets, that have practical significance for portfolio stability.

By comparison, expert benchmarks often exhibited a ‘winner-takes-all’ bias, assigning almost 100% weight to a single asset. Such allocations are vulnerable to idiosyncratic shocks, making them less robust despite occasionally high in-sample performance. Our model’s ability to maintain balanced exposure across assets and adapt allocations to different regimes represents a meaningful step toward more **resilient** and **real-world-ready** portfolio generation.

VI. FUTURE WORK

While this study demonstrates the feasibility of using generative diffusion models for portfolio construction, the results (detailed in the Appendix) suggest that their greatest current strength lies in **structural robustness** rather than raw outperformance on traditional performance metrics.

Several directions could further enhance both performance and practicality:

- 1) **Scaling to larger universes** — Extending the framework to the full S&P 500 or global equity sets would test scalability and improve generalization. This may require hierarchical or factor-based latent representations to manage the expanded asset space efficiently.
- 2) **Incorporating transaction costs and turnover penalties** — Real-world portfolio management must balance

returns with implementation costs. Embedding explicit cost models or soft penalties into the generation process could make the allocations more practical and stable over time.

- 3) **Refined conditioning strategies** — The current temporal conditioning captures broad market regimes, but richer features—such as macroeconomic indicators, sentiment signals, or risk factor exposures—could sharpen the model’s regime adaptability.
- 4) **Performance-oriented fine-tuning** — While the model currently emphasizes diversification, integrating reinforcement learning or policy-gradient fine-tuning could nudge it toward higher-return allocations without sacrificing stability.
- 5) **Benchmark diversity** — Expanding the set of baselines beyond classical optimization and the current expert benchmark could yield a more complete understanding of where generative methods excel and where they lag.

In short, the diffusion approach is promising for building **resilient, constraint-aware portfolios**, but further methodological refinements and richer conditioning may be required to close the gap in pure performance metrics.

REFERENCES

- [1] H. Du, R. Zhang, Y. Liu, J. Wang, Y. Lin, Z. Li, D. Niyato, J. Kang, Z. Xiong, S. Cui, B. Ai, H. Zhou, and D. I. Kim, “Enhancing deep reinforcement learning: A tutorial on generative diffusion models in network optimization,” *IEEE Communications Surveys and Tutorials*, 2024.

APPENDIX



Fig. 2. [Portfolio value] expert guidance from the equal weight portfolio, prices simulated with geometric Brownian motion.

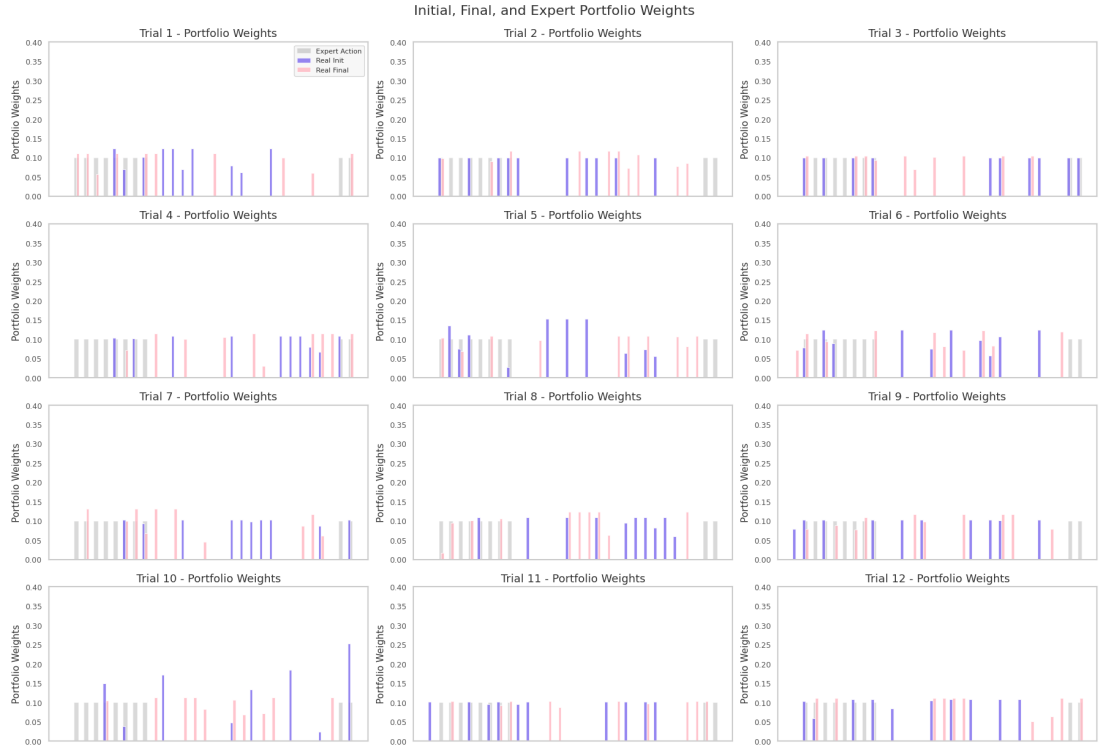


Fig. 3. [Portfolio weights] expert guidance from the equal weight portfolio, prices simulated with geometric Brownian motion.

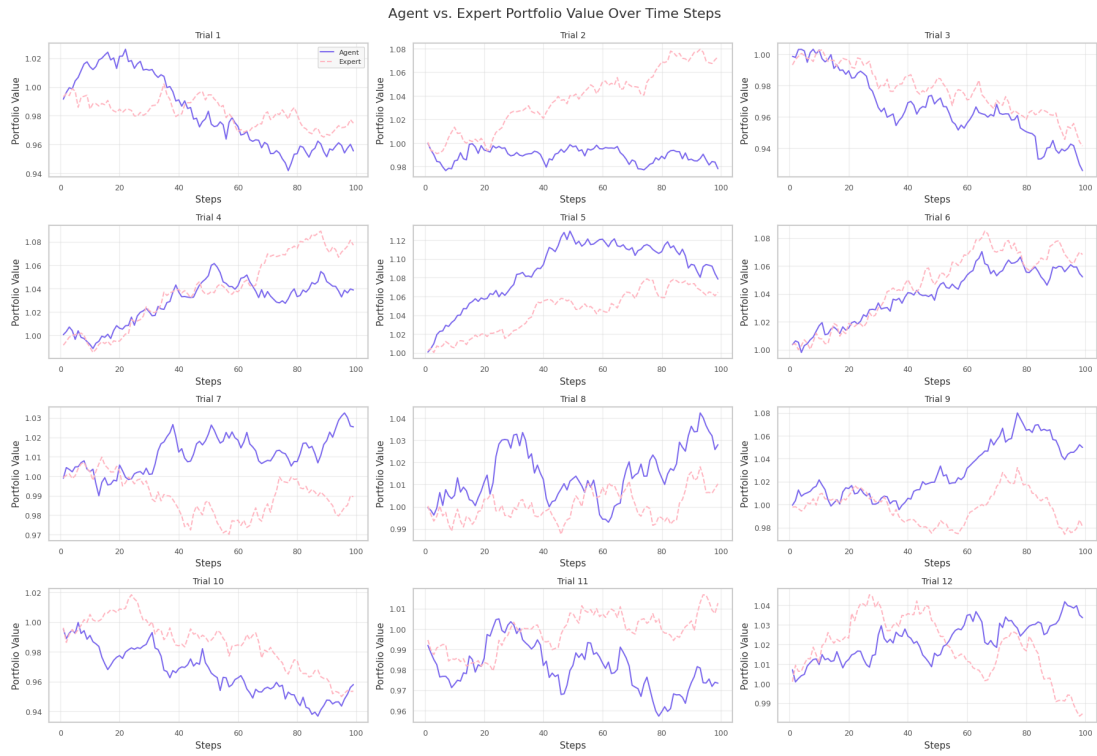


Fig. 4. [Portfolio value] expert guidance from the equal weight portfolio, historical S&P500 prices.

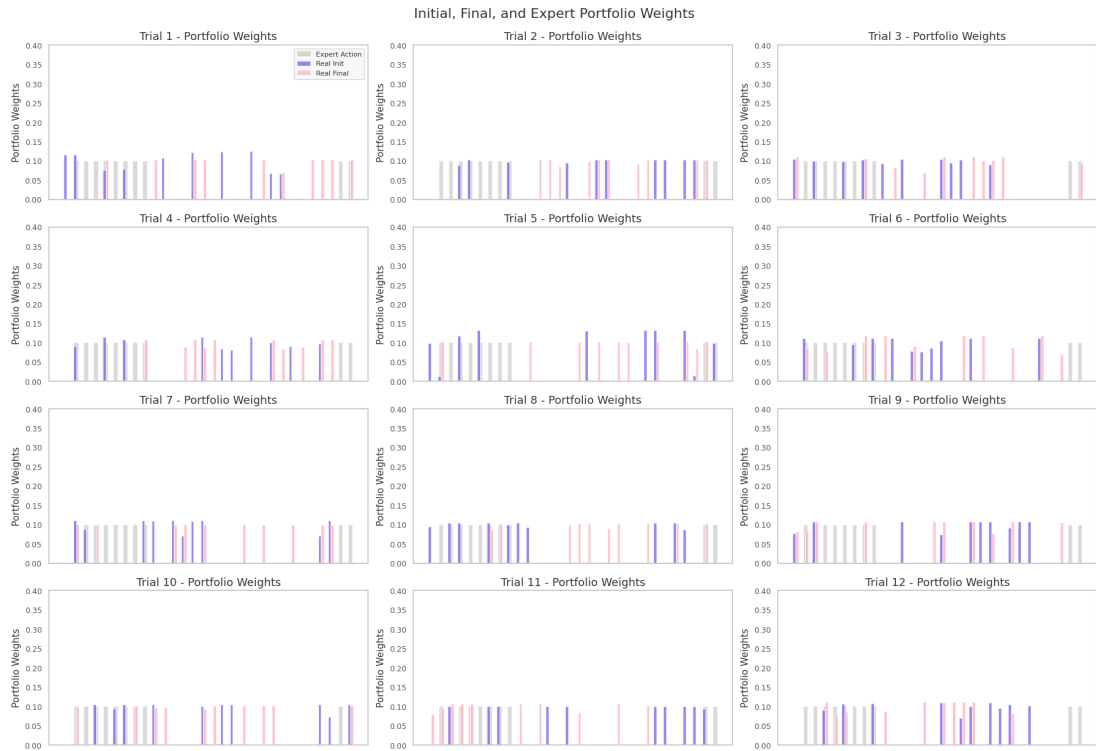


Fig. 5. [Portfolio weights] expert guidance from the equal weight portfolio, historical S&P500 prices.

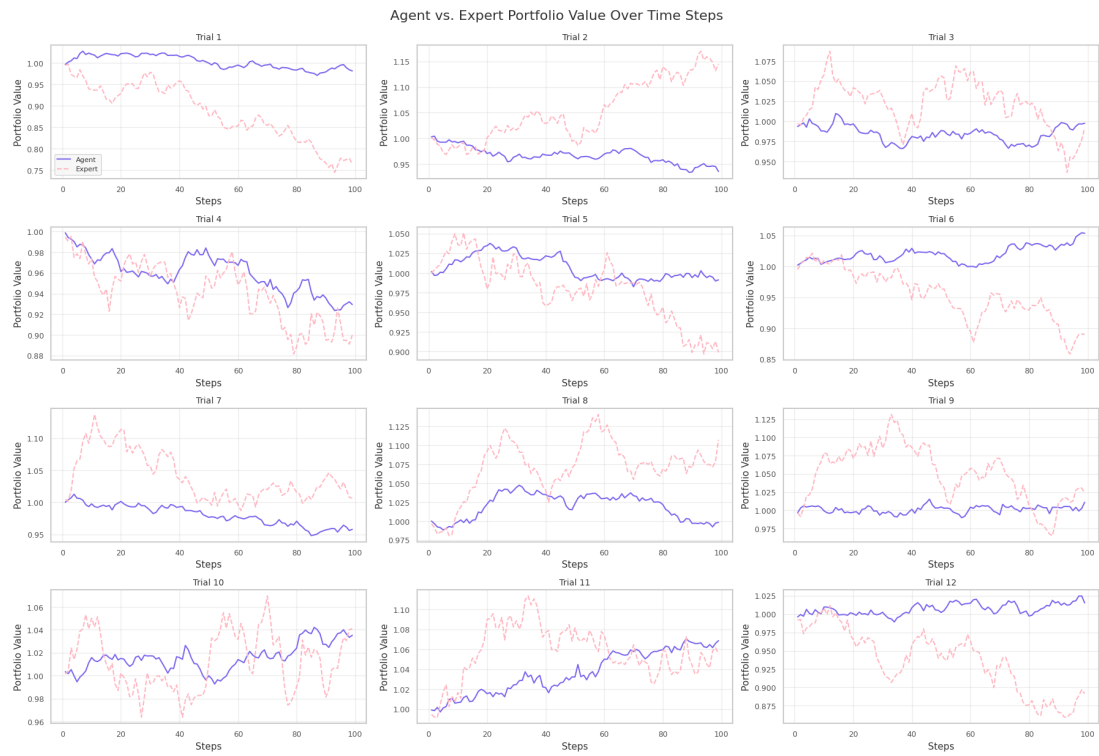


Fig. 6. [Portfolio value] expert guidance from the mean-variance portfolio, prices simulated with geometric Brownian motion.

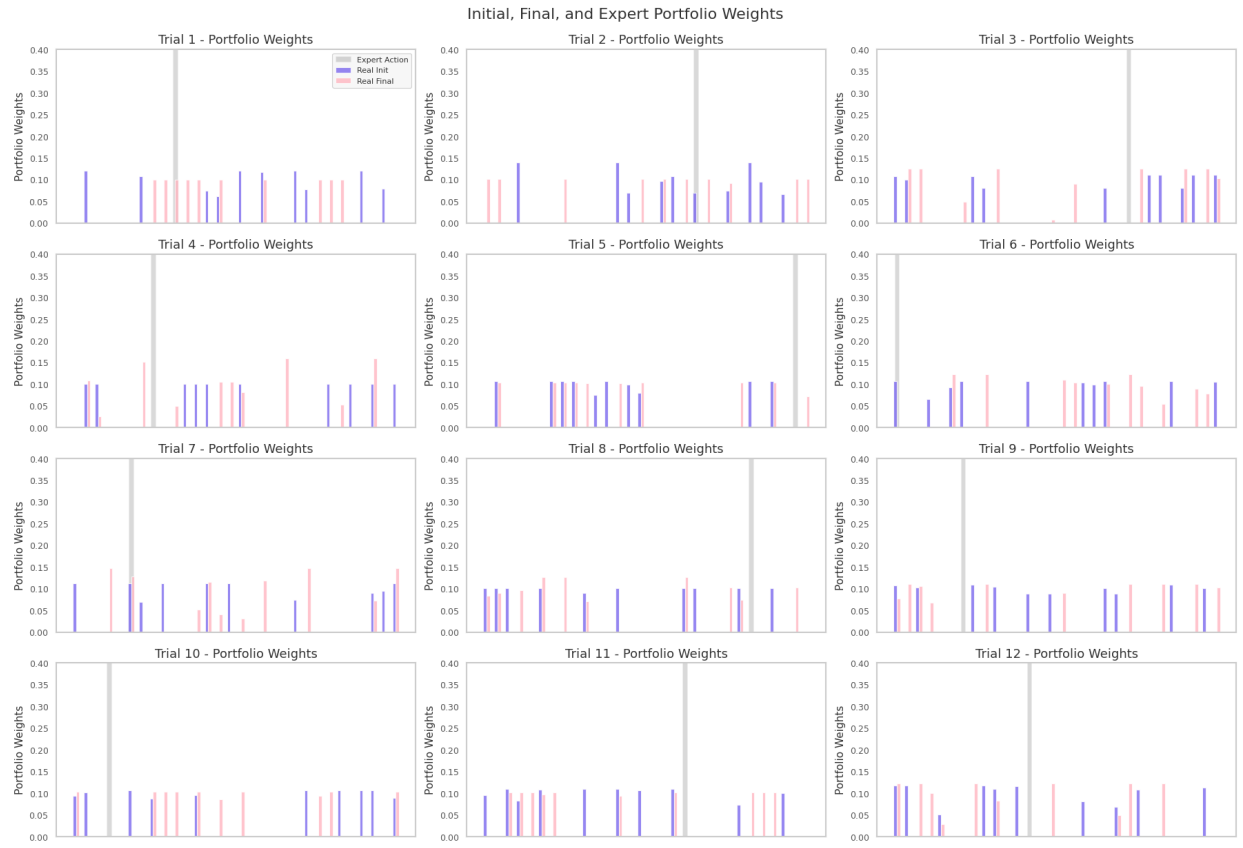


Fig. 7. [Portfolio weights] expert guidance from the mean-variance portfolio, prices simulated with geometric Brownian motion.

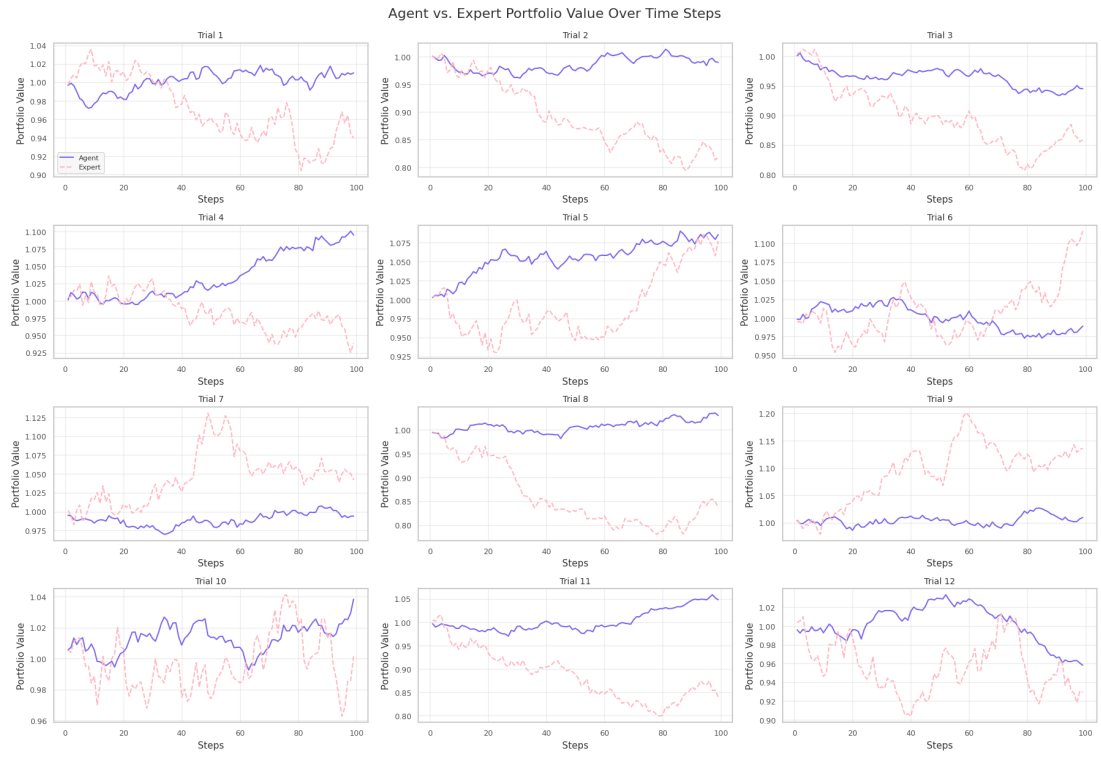


Fig. 8. [Portfolio value] expert guidance from the mean-variance portfolio, historical S&P500 prices.

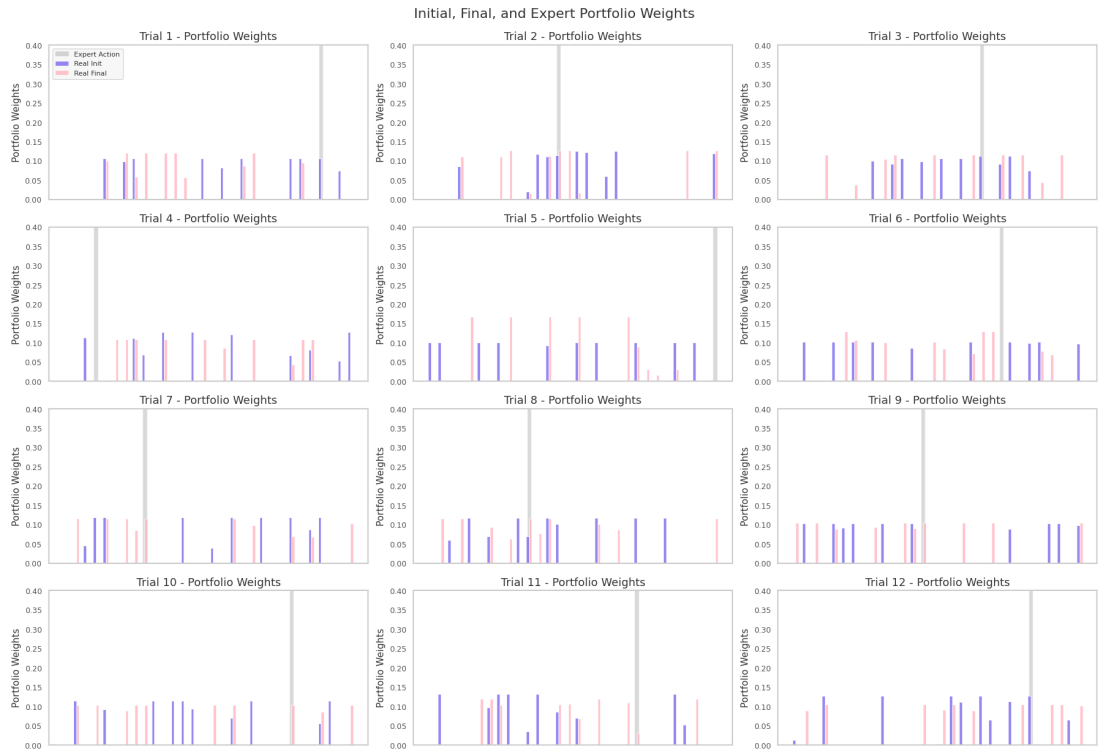


Fig. 9. [Portfolio weights] expert guidance from the mean-variance portfolio, historical S&P500 prices.



Fig. 10. [Portfolio value] expert guidance from the Black-Litterman model, prices simulated with geometric Brownian motion.

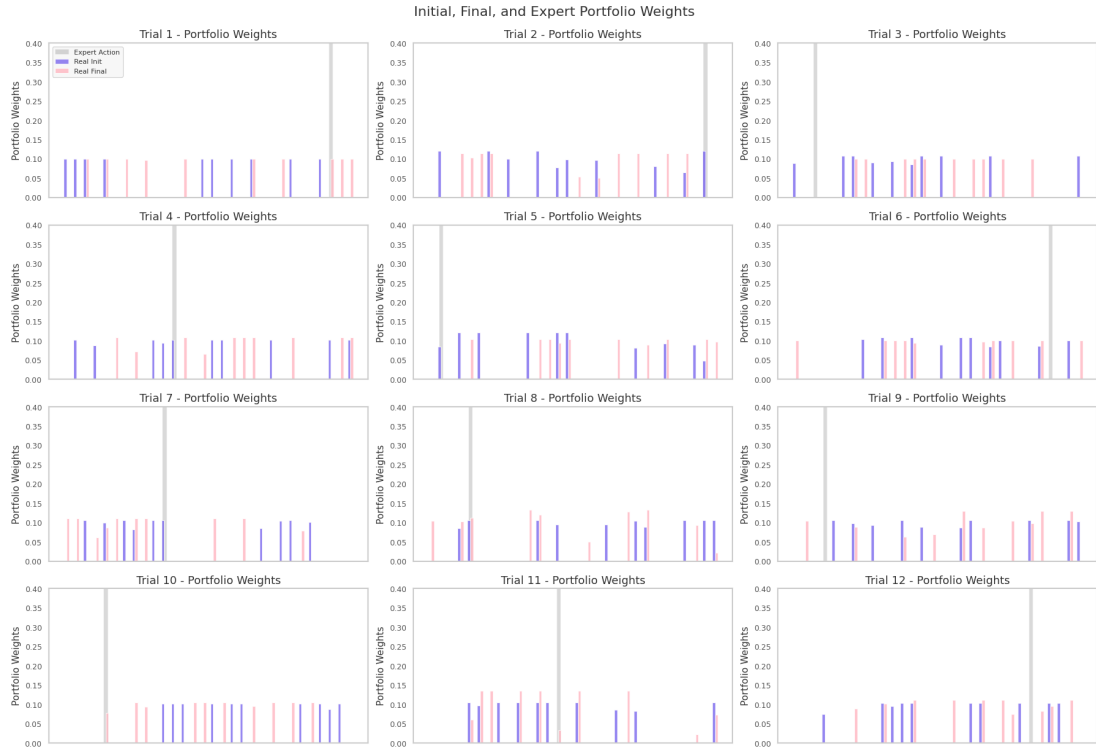


Fig. 11. [Portfolio weights] expert guidance from the Black-Litterman model, prices simulated with geometric Brownian motion.



Fig. 12. [Portfolio value] expert guidance from the Black-Litterman model, historical S&P500 prices.

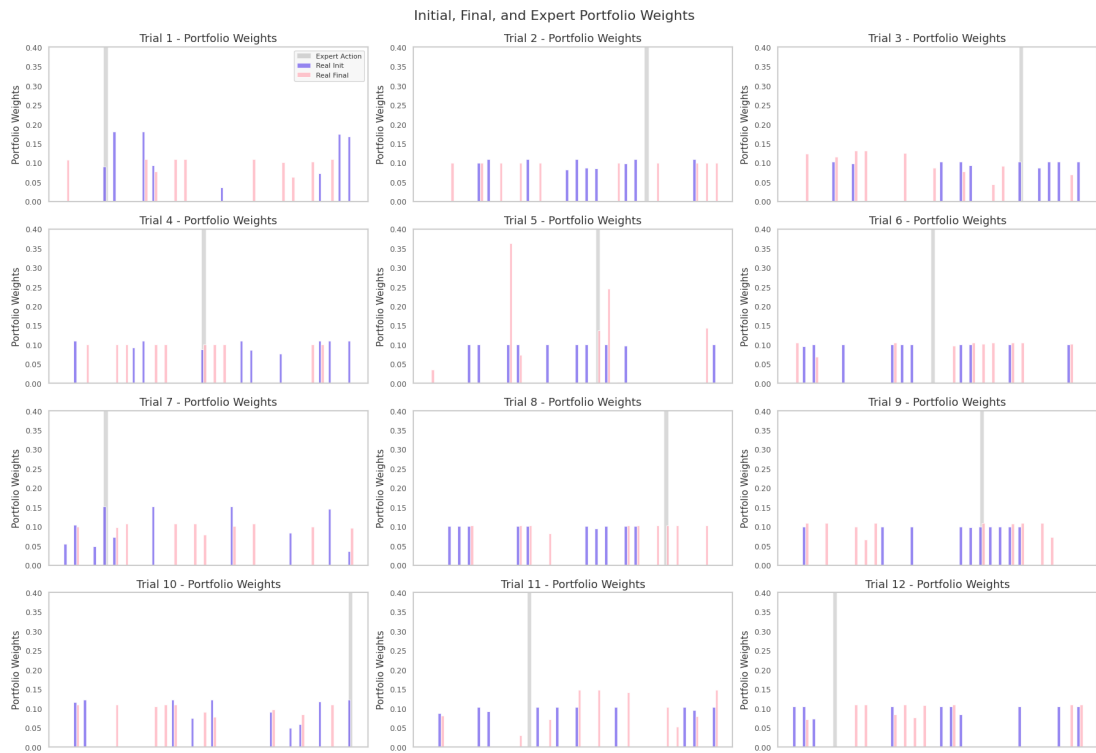


Fig. 13. [Portfolio weights] expert guidance from the Black-Litterman model, historical S&P500 prices.